

Matheus Rolim Sales

Postdoctoral Researcher in Physics

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Research interests

Nonlinear dynamics and chaos; Hamiltonian systems; mechanism of transport and diffusion; classical and quantum billiards; synchronization and chimera states in complex networks; adaptive dynamical networks; neuronal models.

Education

Ph.D. in Physics

Mar 2020 – Oct 2023

State University of Ponta Grossa (UEPG), Brazil

Thesis: *Dynamical aspects of Hamiltonian systems: chaos, stickiness, and recurrence plots*

Advisor: Prof. Dr. José Danilo Szezech Jr. (CAPES)

M.Sc. in Physics

Mar 2018 – Feb 2020

Federal University of Paraná (UFPR), Brazil

Dissertation: *Dinâmica dos níveis de energia do Bilhar de Sinai dessimetrizado e sua relação com bilhares triangulares*

Advisor: Prof. Dr. Marcos Gomes Eleutério da Luz (CNPq)

B.Sc. in Physics

Aug 2017

Federal University of Paraná (UFPR), Brazil

Appointments

Postdoctoral Researcher

Nov 2023 – present

Department of Physics, São Paulo State University (UNESP), Rio Claro, Brazil

Supervisor: Prof. Dr. Edson Denis Leonel (FAPESP)

Research visits

Visiting Fellow

Feb 2025 – Jan 2026

School of Mathematics, Statistics and Actuarial Science, University of Essex, UK

Host: Dr. Chris G. Antonopoulos (FAPESP/BEPE)

Doctoral exchange (PDSE)

Oct 2022 – Mar 2023

Humboldt University of Berlin and Potsdam Institute for Climate Impact Research (PIK), Germany

Hosts: Dr. Serhiy Yanchuk and Prof. Dr. Jürgen Kurths (CAPES/PDSE)

Fellowships and funding

FAPESP Postdoctoral Fellowship (UNESP)

2023 – present

FAPESP Research Internship Abroad, BEPE (University of Essex)

2025 – 2026

CAPES/PDSE Doctoral Exchange (HU Berlin / PIK)

2022 – 2023

CAPES Doctoral Scholarship (UEPG)

2020 – 2023

CNPq Master's Scholarship (UFPR)

2018 – 2020

Software

[dynamicalsys](#), an open-source Python toolkit for the analysis of dynamical systems (NumPy/Numba).

Documentation: dynamicalsys.readthedocs.io.

Publications

Peer-reviewed articles (26), most recent first:

26. M. Mugnaine, **M. R. Sales**, E. D. Leonel, I. L. Caldas, J. D. Szezech Jr., *Role of manifolds in the transport of chaotic trajectories in the standard nontwist map*, [Phys. Rev. E 114, 014222 \(2026\)](#).
25. **M. R. Sales**, E. D. Leonel, C. G. Antonopoulos, *On the dynamical and statistical properties of a quartic mean-field Hamiltonian model*, [Chaos, Solitons & Fractals 211, 118787 \(2026\)](#).
24. **M. R. Sales**, M. Mugnaine, L. C. Souza, I. L. Caldas, E. D. Leonel, J. D. Szezech Jr., *Hierarchical fragmentation of regular islands in a discontinuous nontwist map*, [Chaos 36, 063129 \(2026\)](#).
23. D. Borin, **M. R. Sales**, E. D. Leonel, and D. F. M. de Oliveira, *On the dynamics of the q -Tsallis Gauss Iterated Map*, [Physica A 686, 131352 \(2026\)](#).
22. **M. R. Sales**, E. D. Leonel, and C. G. Antonopoulos, *On the behavior of Linear Dependence, Smaller, and Generalized Alignment Indices in discrete and continuous chaotic systems*, [Chaos, Solitons and Fractals 205, 117884 \(2026\)](#).
21. **M. R. Sales**, L. C. de Souza, D. Borin, M. Mugnaine, J. D. Szezech, R. L. Viana, I. L. Caldas, E. D. Leonel, and C. G. Antonopoulos, *ynamicalsys: A Python toolkit for the analysis of dynamical systems*, [Chaos Solitons and Fractal 201 117269 \(2025\)](#).
20. D. Borin, J. D. Szezech Jr., and **M. R. Sales**, *Characterizing and quantifying weak chaos in fractional dynamics*, [Chaos, Solitons and Fractals 200 117137 \(2025\)](#).
19. L. C. de Souza, **M. R. Sales**, J. D. Szezech Jr., R. L. Viana, I. L. Caldas, and M. S. Baptista, *Pattern formation in symplectic coupled map lattices*, [Chaos, Solitons and Fractals 200 117057 \(2025\)](#).
18. **M. R. Sales**, M. Mugnaine, A. L. R. de Moraes, E. D. Leonel, C. G. Antonopoulos, I. L. Caldas, and J. D. Szezech Jr., *Transport mechanisms associated with non-integer wavenumbers in a discontinuous nontwist map*, [Chaos, Solitons and Fractals 200, 116966 \(2025\)](#).
17. P. G. Buiar, J. D. Szezech Jr., **M. R. Sales**, and G. M. Favero, *Time to focus again on matrix metalloproteinases? Results of complex network analysis involving the pathophysiology of HER2-positive breast cancer*, [ecancer 19, 1850 \(2025\)](#).
16. V. dos Santos, **M. R. Sales**, I. L. Caldas, R. L. Viana, and J. D. Szezech Jr., *Extended networks as a route of stabilization of divergent dynamics*, [Chaos, Solitons and Fractals 193, 116115 \(2025\)](#).
15. **M. R. Sales**, M. Mugnaine, E. D. Leonel, I. L. Caldas, and J. D. Szezech Jr., *Shrinking shrimp-shaped domains and multistability in the dissipative asymmetric kicked rotor map*, [Chaos: An Interdisciplinary Journal of Nonlinear Science 34, 113129 \(2024\)](#).
14. **M. R. Sales**, D. Borin, D. R. da Costa, J. D. Szezech Jr., and E. D. Leonel, *An investigation of escape and scaling properties of a billiard system*, [Chaos: An Interdisciplinary Journal of Nonlinear Science 34, 113122 \(2024\)](#).
13. **M. R. Sales**, D. Borin, L. C. de Souza, J. D. Szezech Jr., R. L. Viana, I. L. Caldas, and E. D. Leonel, *Ratchet current and scaling properties in a nontwist mapping*, [Chaos, Solitons & Fractals 189, 115614 \(2024\)](#).
12. **M. R. Sales**, S. Yanchuk, and J. Kurths, *Recurrent chaotic clustering and slow chaos in adaptive networks*, [Chaos: An Interdisciplinary Journal of Nonlinear Science 34, 063144 \(2024\)](#).
11. A. L. Azevedo, A. C. Maioli, F. Teston, **M. R. Sales**, F. M. Zanetti, and M. G. E. da Luz, *Wave amplitude gain within wedge waveguides through scattering by simple obstacles*, [Phys. Rev. E 109, 025303 \(2024\)](#).
10. L. C. Souza, **M. R. Sales**, M. Mugnaine, J. D. Szezech, I. L. Caldas, and R. L. Viana, *Chaotic escape of impurities and sticky orbits in toroidal plasmas*, [Phys. Rev. E 109, 015202 \(2024\)](#).
9. E. C. Gabrick, **M. R. Sales**, E. Sayari, J. Trobia, E. K. Lenzi, F. S. Borges, J. D. Szezech Jr., K. C. Iarosz, R. L. Viana, I. L. Caldas, and A. M. Batista, *Fractional dynamics and recurrence analysis in cancer model*, [Brazilian Journal of Physics 53, 145 \(2023\)](#).
8. V. dos Santos, **M. R. Sales**, S. S. Muni, J. D. Szezech, A. M. Batista, S. Yanchuk, and J. Kurths, *Identification of single- and double-well coherenceincoherence patterns by the binary distance matrix*, [Communications in Nonlinear Science and Numerical Simulation 125, 107390 \(2023\)](#).
7. **M. R. Sales**, M. Mugnaine, J. Szezech, José D., R. L. Viana, I. L. Caldas, N. Marwan, and J. Kurths, *Stickiness and recurrence plots: An entropy-based approach*, [Chaos: An Interdisciplinary Journal of Nonlinear Science 33, 033140 \(2023\)](#).
6. M. Mugnaine, **M. R. Sales**, J. D. Szezech, and R. L. Viana, *Dynamics, multistability, and crisis analysis of a sine-circle nontwist map*, [Phys. Rev. E 106, 034203 \(2022\)](#).

5. D. R. da Costa, A. Fujita, **M. R. Sales**, J. D. Szezech, and A. M. Batista, *Dynamical properties for a tunable circular to polygonal billiard*, [Brazilian Journal of Physics 52, 75 \(2022\)](#).
4. F. Teston, A. L. Azevedo, **M. R. Sales**, F. M. Zanetti, and M. G. E. da Luz, *The flexibility in choosing distinct Greens functions for the boundary wall method: Waveguides and billiards*, [Journal of Physics A: Mathematical and Theoretical 55, 175201 \(2022\)](#).
3. **M. R. Sales**, M. Mugnaine, R. L. Viana, I. L. Caldas, and J. D. Szezech, *Unpredictability in Hamiltonian systems with a hierarchical phase space*, [Physics Letters A 431, 127991 \(2022\)](#).
2. D. R. da Costa, A. Fujita, A. M. Batista, **M. R. Sales**, and J. D. Szezech Jr, *Conservative generalized bifurcation diagrams and phase space properties for oval-like billiards*, [Chaos, Solitons & Fractals 155, 111707 \(2022\)](#).
1. **M. R. Sales**, A. L. Azevedo, F. Teston, M. G. E. da Luz, and F. M. Zanetti, *Soliton-like structures in the spectrum and the corresponding eigenstates morphology for the quantum desymmetrized Sinai billiard*, [Chaos: An Interdisciplinary Journal of Nonlinear Science 31, 113122 \(2021\)](#).

Book chapter:

- R. L. Viana, L. C. Souza, **M. R. Sales**, M. Mugnaine, J. D. Szezech Jr., I. L. Caldas, N. Marwan, and J. Kurths, *Recurrence-Based Characterization of Stickiness in Hamiltonian Systems*. In: [Recurrence Plots and Their Quantifications: Methodological Breakthroughs and Interdisciplinary Discoveries](#). Springer Proceedings in Complexity, Springer, Cham, 2025.

References

Prof. Dr. Edson Denis Leonel (Postdoctoral supervisor)

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Prof. Dr. José Danilo Szezech Jr. (Ph.D. advisor)

Department of Mathematics and Statistics, UEPG, Brazil · jdsjunior@uepg.br

Dr. Serhiy Yanchuk (Doctoral exchange supervisor)

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Dr. Chris G. Antonopoulos (Postdoctoral host)

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